# Probabilistic Study Graph for BIDS Entity Inference from DICOM Series

## 1 Problem Setting

Given a set of DICOM series extracted from clinical archives and de-identified (e.g., XNAT pipelines), we aim to:

- Infer BIDS-compatible semantic labels at the **series level**.
- Predict a joint label $(d, s)$ where:
  - $d \in \mathcal{D}$ is the BIDS `datatype` (e.g., anat, func, dwi, perf).
  - $s \in \mathcal{S}$ is the BIDS `suffix` (e.g., T1w, bold, dwi, sbref, fieldmap).
- Enforce physics- and protocol-based constraints across series within the same session.

This is formulated as structured prediction using a probabilistic graphical model.

## 2 Series-Level Feature Representation

Each DICOM series $i$ consists of slice-level instances:

$$\mathcal{X}_i = \{x_{i1}, x_{i2}, \dots, x_{iN_i}\}$$

From header metadata, we extract parameter-wise statistics:

- Central tendency (median TR, TE, flip angle, voxel spacing)
- Dispersion (IQR, variance)
- Categorical distributions (ImageType, SequenceName)

- Temporal features (AcquisitionTime distribution)
- Geometry descriptors (orientation matrix, affine consistency)

The resulting engineered feature vector:

$$\mathbf{f}_i \in \mathbb{R}^d$$

A supervised classifier (e.g., XGBoost, Random Forest, structured-text fusion model) outputs:

$$P_{\mathrm{ML}}(y_i \mid \mathbf{f}_i)$$

where $y_i = (d_i, s_i)$ is the joint BIDS label.

**Joint classification is preferred over hierarchical datatype $\rightarrow$ suffix prediction because it prevents error propagation and allows modeling cross-category correlations directly in label space.**

# 3 Probabilistic Study Graph

We define a graph $G = (V, E)$ per session:

- Nodes $V$: DICOM series.
- Edges $E$: relations between series reflecting acquisition logic.

Edges are only constructed within-session to respect study structure.

## 3.1 Rationale from Neuroimaging Literature

Neuroimaging acquisitions exhibit structured dependencies:

- **Fieldmaps** correct susceptibility distortions in EPI.
- **SBRef images** precede BOLD runs for motion correction and co-registration.
- **Diffusion acquisitions** consist of multi-shell or multi-direction sequences sharing geometry.
- **Anatomical scans** provide spatial reference for functional normalization.

These relationships motivate structured modeling rather than independent classification.

## 3.2 Edge Types and Scientific Justification

### 1. Same-Session Membership

All nodes in a session form a connected component:

$$(i, j) \in E_{\text{session}}$$

Rationale:

- Ensures inference operates within coherent acquisition contexts.
- Prevents cross-subject contamination.

### 2. Temporal Adjacency

Define median acquisition timestamp $t_i$ per series.

Edges:

$$(i, j) \in E_{\text{time}} \quad \text{if} \quad |t_i - t_j| < \delta$$

Rationale:

- SBRef typically precedes BOLD.
- Fieldmaps are often acquired immediately before/after target runs.

### 3. Geometry Similarity

Define affine matrices $A_i$ per series.

Similarity score:

$$S_{ij}^{\text{geom}} = \exp(-\|A_i - A_j\|_F)$$

Rationale:

- Fieldmaps and BOLD share slice prescription.
- SBRef and BOLD share identical geometry.

### 4. Fieldmap–Target Pairing

Based on:

- PhaseEncodingDirection
- EchoTime
- TotalReadoutTime

Encodes compatibility constraints:

$$\phi_{ij}^{\text{fmap}}(y_i, y_j)$$

**5. SBRef–BOLD Coupling**

Encodes strong positive compatibility:

$$\phi_{ij}^{\text{sbref}}(y_i = \text{sbref}, y_j = \text{bold}) \gg 1$$

**6. Mutual Exclusion**

Prevents impossible combinations within session:

$$\phi_{ij}^{\text{excl}}(y_i, y_j) \approx 0$$

Examples:

- Two T1w series rarely share identical geometry and timestamp.
- Derived/localizer images incompatible with primary datatypes.

# 4 Markov Random Field Formulation

Let $Y = \{y_i\}$ denote all labels.

We define an undirected MRF:

$$P(Y) = \frac{1}{Z} \prod_{i \in V} \psi_i(y_i) \prod_{(i,j) \in E} \psi_{ij}(y_i, y_j)$$

## 4.1 Node Potentials

Derived from classifier:

$$\psi_i(y_i) = P_{\text{ML}}(y_i \mid \mathbf{f}_i)$$

## 4.2 Pairwise Potentials

For each edge type $k$:

$$\psi_{ij}(y_i, y_j) = \prod_k \exp\left(\lambda_k \cdot g_k(y_i, y_j, \mathbf{f}_i, \mathbf{f}_j)\right)$$

where:

- $g_k$ encodes compatibility.
- $\lambda_k$ controls strength.

### 4.3 Partition Function

$$Z = \sum_{Y} \prod_{i} \psi_i(y_i) \prod_{(i,j)} \psi_{ij}(y_i, y_j)$$

Exact computation is intractable for loopy graphs.

## 5 Inference via Loopy Belief Propagation

We use Loopy Belief Propagation (LBP):

- Iterative message passing algorithm.
- Approximates marginal distributions in cyclic graphs.
- Empirically effective for pairwise MRFs.

Message update:

$$m_{i\to j}(y_j) = \sum_{y_i} \psi_i(y_i)\psi_{ij}(y_i, y_j) \prod_{k\in N(i)\setminus j} m_{k\to i}(y_i)$$

Belief:

$$b_i(y_i) \propto \psi_i(y_i) \prod_{k\in N(i)} m_{k\to i}(y_i)$$

Final label:

$$\hat{y}_i = \arg\max_{y_i} b_i(y_i)$$

### 5.1 Why LBP?

- Efficient for moderate-sized session graphs.
- No need for differentiable potentials.
- Avoids heavy computation of MCMC.
- Structured mean-field often underestimates multimodal interactions.

## 6 Assignment of Additional BIDS Entities

After semantic label inference:

- Run index assigned by clustering temporally ordered identical-label series.
- Fieldmap IntendedFor determined via posterior compatibility scores.
- Echo, direction, acquisition labels derived from DICOM tags.

Not all BIDS entities are required for categorization — only sufficient disambiguation is enforced.

## 7 Core Conceptual Summary

This framework combines discriminative learning with structured probabilistic reasoning. Local evidence from DICOM metadata produces candidate semantic labels. A session-level MRF encodes acquisition physics, protocol regularities, and mutual exclusivity constraints. Approximate inference via LBP refines predictions to produce globally coherent BIDS annotations.